

Tasks: propose, design, implement, and document software component.

The report of software component development must contain:

1. Component description:
 - a. provided functions,
 - b. purpose,
 - c. business problems that component solves,
 - d. intended use,
 - e. restrictions for component usage (technical, business),
 - f. other important information for component users.
2. Component architecture (internal architecture that is not provided to end-users):
 - a. design component with an easy maintenance in mind,
 - b. class diagram,
 - c. usage scenarios (sequence diagrams of predicted component usage in end user scenarios),
 - d. state diagrams,
 - e. deployment diagrams (how the component must be deployed),
 - f. additional information and models that reveal inner workings of the component.
3. Component description (documentation that is provided to component users):
 - a. class diagrams,
 - b. detailed class, methods, and packages descriptions,
 - c. api documentation (ex.: javadocs, doxygen documentation that is generated from code annotations),
 - d. usage examples – how to use the component:
 - i. code samples with comments that present typical component usage,
 - ii. samples should allow end-user to get start using the component in 5 mins,
 - iii. samples allowing end-users to make a decision if component is suitable for their projects.
 - e. user guide,
 - f. installation instructions and dependencies list (ex.: required additional components),
 - g. requirements for deployment environment,
 - h. changes log,
 - i. license.
4. Perform analysis of similar, competing components:
 - a. do research on commercial components presentation and documentation,
 - b. compare how open-source components are presented, distributed, documented,
 - c. research how components are documented,
 - d. based on research findings improve your component description, documentation.